**RAJIV GANDHI COLLEGE OF ENGINEERING AND TECHNOLOGY**
**MODEL EXAMINATION (2022-23) ODD SEM**
**ENGINEERING THERMODYNAMICS**

**SECTION: A, B& C** **MAX.MARKS: 75**
**DATE: 13.02.2023**

**PART – A** **(10 x 2 = 20)**

**ANSWER ALL THE QUESTIONS**

1. Define path and process in Thermodynamics.
2. What is the difference between classical and statistical approaches in Thermodynamics?
3. Define heat engine.
4. Define Clausis- statement.
5. What is PMM 2?
6. State Carnot’s theorem.
7. Write the expression for air standard efficiency of dual cycle?
8. What is reversible process and irreversible process?
9. Name any four commonly used refrigerants.
10. Write down the advantages of vapour compression refrigeration system?

**PART-B** **(5 X 11 = 55)**

**ANSWER ALL THE QUESTIONS**

11. If a gas of volume 6000 $cm^3$ and at a pressure of 100 kPa is compressed quasi statically according to $pv^2$ =constant until the volume become 2000 $cm^3$. Determine the final pressure and the work transfer.

(OR)

12. Discuss quasi-static process with neat diagram and plot the quasi-static process in a P-V plot.

13. A quantity of air having a volume of 0.04 $m^3$ at a temperature of 525 K and a pressure of 150 $N/cm^2$ is expanded at constant pressure to 0.08 $m^3$it is then expanded adiabatically to 0.12 $m^3$.
Find: (a)Temperature and pressure at the end of the adiabatic process
(b)Work done during each stage

(OR)

14. A steady flow thermodynamic system receives fluid at the rate of 6 Kg/min with an initial pressure of 2 bar, velocity, 150 m/s, internal energy 800 Kj/Kg and density 27 Kg/m$^3$. The fluid leaves the system with a final pressure of 8 bar, velocity 200 m/s, internal energy 800 Kj/Kg and density 5 Kg/m$^3$. If the fluid receives 80 Kj/Kg of heat during passing through the system and rises through 60 m, determine the work done during the process.

15. A reversible heat engine operates between two reservoirs at temperatures 700°C and 50°C. The enginedrives a reversible refrigerator which operates between reservoirs at temperatures of 50°C and – 25°C. The heat transfer to the engine is 2500 kJ and the net work output of the combined engine refrigerator plant is 400 kJ. (i) Determine the heat transfer to the refrigerant and the net heat transfer to the reservoir at 50°C.(ii) Reconsider (i) given that the efficiency of the heat engine and the C.O.P. of the refrigerator are each 45 per cent of their maximum possible values.

(OR)

16. A reversible heat engine between a source at 972$^\circ$ C and two sinks one at 127$^\circ$ C and another at 27$^\circ$ C. The energy rejected is same at the both sinks. What is the ratio of heat supplied to the heat rejected? Also calculate the efficiency

17. Derive the efficiency of diesel cycle and write the formula of mean effective pressure?

(OR)

18. The minimum pressure and temperature in an Otto cycle are 100 kPa and 27°C. The amount of heat added to the air per cycle is 1500 kJ/kg. Determine, (a) the pressure and temperature at all points of the air standard Otto cycle, (b) Specific work and (c) Thermal efficiency of the cycle for compression ratio of 8:1. Take for air, $C_P$ = 0.72 kJ/kg K and $\gamma$ = 1.4.Assume that compression ratio is 15 and working fluid is air. Consider all conditions to be ideal.

19. Explain with neat sketch the working of Linde Hampson System for Liquefaction of Air.

(OR)

20. Explain the Vapour Absorption Refrigeration System with neat sketch.